PAPER ID-311734

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
AUTOMOTIVE CHASSIS AND SUSPENSION

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Differentiate between a Live Axle and a Dead Axle.	CO-1	K1
b.	State the need for a clutch in an automobile.	CO-2	K1
c.	List four different types of loads acting on an automotive frame.	CO-3	K1
d.	State the difference betweenundersteering and oversteering.	CO-4	K1
e.	Briefly explain the purpose of a differential lock.	CO-5	K1
f.	Define the term Wheel Balancing and state its necessity.	CO-1	K1
g.	Name the key components of a Hydraulic Braking System.	CO-2	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

Q no.	Question	CO	Level
a.	Discuss the advantages and disadvantages of front wheel drive vehicles over rear-wheel-drive vehicles.	CO-1	K3
b.	Draw neat diagram of a constant mesh gear box and explain its working.	CO-2	K3
c.	Draw a neat sketch of the air brake system of a motor vehicle and describe how it works.	CO-3	K3
d.	Describe, with a neat sketch, the working of an Electric power steering system.	CO-4	K3
e.	State the functions and requirements of an automobile wheel tyre.	CO-5	K3

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	The frames are narrower at their fronts than at rear- why? why are the frames heavier in front than in rear in case of vehicles equipped with independent front-wheel suspension?	CO-1	K3
b.	Describe frameless construction of car bodies. How does it fifer from unitary construction.	CO-1	K3

4. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Draw neat sketch of centrifugal clutch, and explain the construction and working of this clutch.	CO-2	K3
b.	Draw neat sketch of a cone clutch, and explain the construction and working of this clutch.	CO-2	K3

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5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Briefly explain the need for a suspension system and describe the working of a hydraulic shock absorber.	CO-3	K3
b.	Compare the constructional and operational aspects of disc and drum brakes. Which one is preferred and why?	CO-3	K3

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	State and explain the principle of Davis steering mechanism.	CO-4	K3
b.	Describe, with a neat sketch, the working of a Linkage power steering system.	CO-4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Discuss the constructional details and function of different types of wheels used in automobiles. Explain the process of Wheel Alignment.	CO-5	K3
b.	Explain, with the help of a suitable sketch, the constructional aspects of a disc type wheel.	CO-5	K3

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